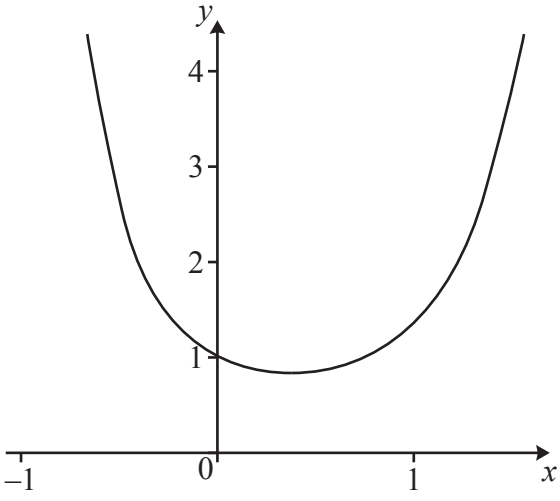
10 The diagram shows part of the graph of the function $y = \dfrac{e^{x^2}}{x+1}$ which is defined for $x > -1$.



(a) Find an expression for $\dfrac{dy}{dx}$. **[3]**

(b) Determine the range of values of x where the gradient of the function is negative. **[3]**

(c) Use the trapezium rule with 4 strips to estimate the area of the region bounded by the curve, the axes and the line $x = 1$. **[3]**

(d) Determine whether the estimate in part **(c)** is an under- or over-estimate. Give a reason for your answer. **[1]**